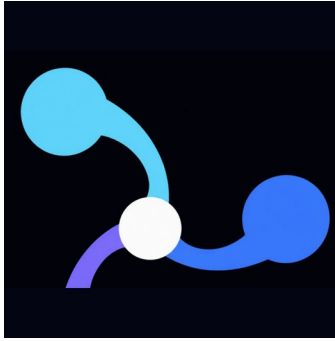




Knowledge Ball

A spatial knowledge network for claims that can be inspected,
challenged, revised, and learned

Knowledge is energy.

Public Project Edition • English

White Paper 1.0 • Public Draft

August 2026

Status and interpretation

This paper sets out the product direction, protocol design, and engineering roadmap for Knowledge Ball. The project remains at the prototype and protocol-convergence stage. Implemented capabilities, committed designs, and research directions are identified separately throughout.

Status	Meaning
Implemented	Running code or a testable workflow exists; this does not imply production-scale validation.
Committed design	The rule is part of the project's current specification but still requires full end-to-end engineering.
Research direction	A longer-term possibility, not a current feature or delivery commitment.

Executive thesis

The web made publishing cheap. Search made retrieval fast. Generative AI made answers nearly instantaneous. None of those shifts solved a more basic problem: most people still cannot inspect the structure behind an answer. The premises are scattered, counterarguments are buried in comment streams, revisions overwrite earlier reasoning, and personal confidence is routinely confused with public validity.

Knowledge Ball is a spatial, versioned knowledge network built around three separations: an enduring knowledge identity is not the same thing as a particular claim version; a public verdict is not the same thing as an individual's mastery; and a fluent model output is not the same thing as accepted knowledge. Claims, reasoning, evidence, challenges, verdicts, and learning state are represented as distinct objects with explicit relationships.

The three-dimensional interface is an operational view of that protocol, not a decorative graph. Semantic status determines shells, argument structure determines chains, deterministic placement preserves spatial memory, and level-of-detail rules keep the field usable on mobile devices. The project's goal is not to manufacture certainty. It is to make knowledge easier to audit, harder to rewrite silently, and more useful for both people and AI systems.

What this paper commits to

- Knowledge Ball treats claims and arguments as first-class data, rather than flattening them into pages, posts, or notes.
- Verdicts are provisional protocol outcomes. A new version can replace the active view without erasing the history that led there.

- A candidate rejected during initial validation never becomes public knowledge; only the corresponding accounting settlement remains. Accepted claim versions retain an auditable lineage.
- Energy is conserved through double-entry accounting. Balances are projections of the ledger, and clients never have authority to mutate them directly.
- The spatial model uses semantic shells, 5R spacing, Fibonacci directions, FCC placement, and canonical spherical addresses to keep geometry stable and reconstructable.
- The current baseline is a TypeScript/Vite/Three.js web application shared with Android and Windows packaging. Protocol closure, mobile stability, and deterministic layout are the near-term engineering priorities.

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1. The thesis

Thesis The next useful layer of the knowledge stack is not another answer engine. It is an inspectable substrate that shows how an answer is formed, contested, and updated.

Most digital knowledge products optimize one stage of the journey. Search engines locate documents. Encyclopedias curate summaries. Social platforms distribute opinions. Note-taking tools externalize memory. Language models synthesize responses. The user is still responsible for reconstructing the dependency graph: which definition is being used, which premises matter, what evidence is independent, which objection is live, and whether a revision invalidates an earlier conclusion.

Knowledge Ball starts below the interface. It defines a minimum set of objects and transitions that can support public review, private learning, machine retrieval, and a spatial view without allowing any one layer to corrupt the others. The product succeeds only if it lowers the cost of understanding structure, not if it merely renders a larger graph.

1.1 The user promise

- See the active conclusion without losing access to dissent and history.
- Trace a claim to its premises, reasoning, sources, and governing version.
- Separate “the network currently accepts this” from “I understand or believe this.”
- Contribute a precise claim or argument instead of adding another unstructured comment.
- Return to a stable spatial map rather than relearning a shuffled graph every session.

2. What Knowledge Ball is, and is not

2.1 Product and protocol

Knowledge Ball is both an end-user product and a protocol boundary. The product provides a searchable 3D field, claim detail views, history, contribution workflows, and a private mastery mode. The protocol defines identities, versions, reasoning membership, eligibility, staking, adjudication, settlement, and authoritative events. The interface may evolve quickly; the protocol is expected to change deliberately and visibly.

2.2 Explicit non-goals

- It is not a popularity oracle. Votes produce a bounded, revisable verdict under defined eligibility and stake rules.
- It is not a neutral pile of incompatible views. The main view must identify the active version while preserving structured opposition.

- It is not a 3D visualizer in search of a use case. Geometry must encode semantics, dependencies, persistence, or interaction value.

The governing metric is therefore not total nodes. It is whether a person can answer, with less effort: What is this? What does it depend on? Why is it accepted? What is the strongest live challenge? What do I personally know about it?

3. The core primitives

Knowledge systems often use one record for both an enduring topic and its current wording. That shortcut breaks citations, revision history, and dispute handling. Knowledge Ball separates identity from assertion and treats reasoning as an object with its own integrity constraints.

Primitive	Purpose	Invariant
KnowledgeNode	A stable identity and reference point	Keeps a durable ID while the active claim version changes.
KnowledgeClaimVersion	A specific assertion at a point in time	Material semantic change creates a new version; prior text is not silently overwritten.
ReasoningNode	An explicit inference connecting premises to one concrete conclusion	Conclusion identity and premise membership are persisted, not inferred from temporary screen edges.
Relation	Typed premise, conclusion, dependency, or special linkage	Business meaning is explicit and independently validated.
Verdict	The protocol's current outcome for one claim version	PENDING, CORRECT, or INCORRECT is version-scoped and reversible.
PersonalMastery	A person's private learning and visibility state	Never enters public voting, public event streams, or public accuracy metrics.

3.1 Names are labels, not identity

The same phrase may refer to an active claim, a historical formulation, an opposing branch, or a different context. Stable IDs and graph position carry identity. Where same-name variants appear in one chain, the interface adds contextual version tags; the dominant node retains the base label. Those tags are navigation aids, not part of the claim's meaning.

3.2 Arguments must terminate in a claim

A reasoning node cannot float free of a concrete conclusion. If its conclusion is hidden, unavailable, pending, or excluded by the current personal view, the reasoning must also be hidden. The default

scene shows the winning path. Superseded and losing paths remain inspectable in history and detail views without overwhelming the primary field.

4. From submission to public knowledge



Figure 1. Candidate lifecycle. The numbered stages are described below.

1. **Submit.** An eligible, verified user supplies the claim, description, sources, and graph relationships, and locks 1 unit of Energy.
2. **Preflight.** The server checks normalized names, reference integrity, relation legality, cycle constraints, duplicate risk, and whether the edit is actually a new semantic version.
3. **Validate.** A passing submission enters a candidate space, separate from the public knowledge graph. Support and opposition each stake 1 unit and are bound to that exact version.
4. **Adjudicate.** The first side to reach the current eligibility-adjusted threshold produces the active verdict. The present parameter sequence grows in steps of 1, 2, 4, 8, and so on; population bands are versioned protocol parameters.
5. **Settle.** Verdict, ledger transfers, lock release, and the current projection are committed in one server-side transaction. Partial success is not permitted.
6. **Admit or reject.** A successful first-round version enters the public graph. A failed first-round candidate is removed from the knowledge surface; a non-content settlement record remains for accounting integrity.
7. **Revise.** Once admitted, a claim is changed through a new version or an explicit opposing branch. If the challenger becomes dominant, the active projection changes while the accepted lineage remains auditable.

Hard rule Creation is not acceptance. Every new or materially changed claim must pass versioning, stake, adjudication, and atomic commit.

4.1 A verdict is a protocol fact, not an eternal truth

The word CORRECT means that a version is currently accepted under a known evidence set, voter eligibility rule, threshold, and settlement policy. That is useful precisely because it is bounded. The system can say what is active now while preserving the mechanism for reversal. Reliability must ultimately combine source quality, argument quality, domain relevance, historical calibration, and resistance to coordinated manipulation, not raw vote count.

4.2 Local and global eligibility

The protocol supports a local scope built around the relevant knowledge neighborhood, including a ten-hop eligibility boundary, and a global scope for foundational or cross-domain claims. The exact eligibility function remains an engineering and governance problem: it must reduce drive-by voting without turning expertise into a closed guild.

5. Energy without token economics

Purpose Energy prices actions and records judgment outcomes. It is designed to create accountability inside the protocol, not financial value outside it.

Mechanism	Current rule	Design intent
Starting balance	0; current lower bound is -10	Avoid a tradable launch allocation while allowing limited early participation.
Submission stake	1 unit per candidate version	Place a cost on spam, duplication, and careless submissions.
Judgment stake	1 unit per support or oppose action, version-bound	Make the object of judgment unambiguous.
Settlement	The winning side receives the losing stake under the protocol's distribution rule	Close the loop between judgment and consequence.
Conservation	Every internal transfer balances; global net Energy remains zero	Prevent arbitrary minting, silent burning, and client-side balance claims.
Precision	0.000001	Make multi-party allocation and rounding deterministic.

The ledger is the source of truth; a balance is only its projection. Clients request actions and display outcomes. They do not submit authoritative balances, choose the actor ID, or invoke privileged settlement functions. Server-side constraints, idempotency keys, and atomic transactions are required for every lock, release, distribution, and reversal.

5.1 Incentive failure modes

- Winner-takes-opponent-stake dynamics can reward herding. Delayed revelation, domain eligibility, or calibration-weighted influence may be necessary.
- A negative balance floor supports onboarding but can subsidize disposable accounts unless paired with device risk, rate limits, and cooldowns.

6. Two graphs, one person: public validity and private mastery

A knowledge network should not confuse what the community currently accepts with what an individual has learned. Knowledge Ball keeps those questions in separate stores and separate permission domains.

Dimension	Public knowledge	Private mastery
Audience	Authorized viewers of the public space	The individual, or a private space they explicitly authorize
State	Versions, verdicts, sources, arguments, stakes, history	Mastery, illuminated/learned state, saved state, hidden state, learning progress
Affects adjudication	Yes, through protocol actions	No
Event stream	Public protocol stream	Separate private persistence; excluded from public accuracy and voting
Control	Subject to audit and governance rules	Exportable, resettable, and deletable by the user

The separation also improves rendering. When a conclusion is hidden in personal mode, its argument disappears with it. Pending or disputed conclusions can be hard-gated. The user receives a coherent personal map without rewriting the shared graph or leaving orphaned reasoning in the scene.

7. Why the interface is spatial

Interaction model Overview first. Relationships second. Full content on zoom or focus. Spatial memory should survive ordinary updates.

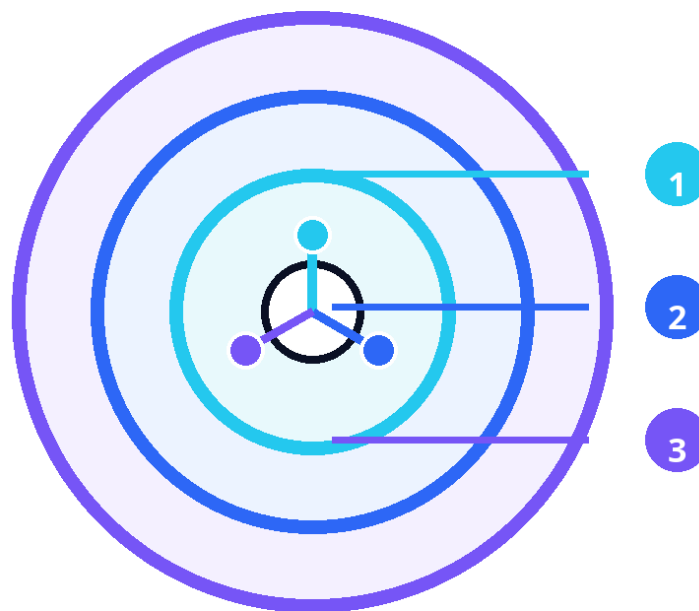


Figure 2. The semantic shells and fixed system core.

7.1 Semantic shells

- Core sun: three code-defined system anchors used for navigation and presentation. They do not enter the database, event stream, voting, Energy, editing, negation, or prerequisite graph.
- Cyan inner shell: definitions, facts, and logic symbols that ground other knowledge.
- Blue middle shell: other admitted public knowledge.
- Violet outer shell: pending, suspended, and disputed content, plus hypotheses, predictions, opinions, and value claims.

7.2 Deterministic geometry

R is the diameter of a knowledge ball. The target and exclusion scale between knowledge nodes is 5R. Neighboring shells and outward capacity expansion also move in minimum 5R increments. The layout solves connected components and difficult chains first, optimizes each chain's internal branches before global placement, then freezes that local geometry.

Fibonacci Sphere samples provide candidate directions. An FCC lattice at R-scale resolution provides snapping and de-duplication. Occupying one knowledge position also reserves all lattice candidates within 5R. Reasoning nodes are temporarily compressed out of the occupancy graph, with premise and conclusion membership retained, then restored after knowledge placement so they cannot distort capacity.

7.3 Canonical addresses

The spherical address system follows an icosahedral hex-grid approach. Generation may use (shell, face, i, j, k), but persistence uses only (shellID, cellID). Shared edges and vertices between parent triangles are canonicalized to one cellID. A physical point therefore has one authoritative address even when multiple construction faces can describe it.

7.4 Mobile legibility

The camera prioritizes the nearest visible shell, hides rear-facing labels, and maintains a bounded label budget. Zoom, rotation, and focus change presentation without changing semantic addresses. A chain-focus interaction can draw the selected component into the center while collapsing unrelated nodes, then restore the prior scene through the inverse transition.

8. Authoritative data flow

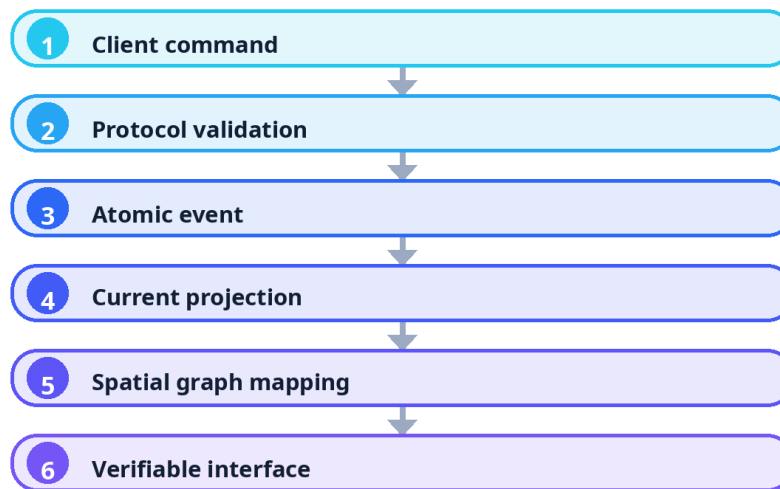


Figure 3. Commands flow through protocol enforcement before events, projections, and the spatial client are updated.

Layer	Responsibility	Authority boundary
Product client	Browse, search, contribute, inspect history, manage private mastery	Requests operations; does not author verdicts, balances, or canonical events.
3D graph engine	Map the projected graph to shells, chains, labels, and camera behavior	Controls geometry only; never changes knowledge semantics.
Protocol engine	Validate commands, versions, relations, eligibility, thresholds, and settlement	Rules are independently testable and parameter-versioned.
Events and projections	Record accepted state changes and build current query views	Events append; projections rebuild; rejected first-round candidates leave settlement only.

Layer	Responsibility	Authority boundary
Identity and sync	Stable user identity, authentication methods, devices, and cross-client state	Login methods may change without changing user history.
Database and security	Postgres/Supabase, row-level security, atomic functions, backups, audit	The server is authoritative; privileged operations never ship to clients.

8.1 Verifiable history without inventing a blockchain

Accepted public state changes are moving toward an append-only event model with rebuildable projections. Stable IDs, previous-event hashes, and content digests can expose silent modification. Independent replicas improve recovery. A future batch of events may be summarized by a Merkle root and anchored to a public network, but that proves that a set of events existed at a time; it does not outsource knowledge adjudication to blockchain consensus.

8.2 Current technical baseline

The current client is built with TypeScript, Vite, and Three.js. The web application is the shared implementation used for Android and Windows packaging. The backend direction uses Supabase/Postgres for authentication, event streams, the Energy ledger, and row-level security. This baseline supports fast iteration; production readiness still depends on idempotent commands, concurrency tests, migration rollback, offline recovery, observability, backup restoration, and sustained mobile performance.

9. Identity, privacy, and security

- Anonymous visitors may inspect public knowledge. Contribution, voting, and public state changes require authentication and protocol eligibility.
- A stable internal user_id preserves account and ledger history while phones, passwords, devices, and future passkeys can change.
- The server derives actor_id from the authenticated session. A client cannot nominate a different author.
- Private mastery lives behind separate tables, policies, and least-privilege access; it does not feed public events or accuracy scores.
- Authenticated users must not receive execute access to conservation, settlement, or other privileged database functions outside the protocol path.

Threat	Failure	Primary controls
Knowledge poisoning	Coordinated promotion of a false claim	Version-bound action, evidence requirements, domain eligibility, structured opposition, reversibility.
Private-state leakage	Exposure of learning gaps, interests, or beliefs	Physical separation, private-by-default policy, export/delete controls, access auditing.

10. Product paths that can create real value

Knowledge Ball should not begin by trying to structure the entire internet. The practical wedge is a domain where prerequisites matter, sources can be checked, and users are already paying the cost of reconstructing arguments manually.

10.1 Priority use cases

- Learning maps: courses, concepts, definitions, theorems, and prerequisites arranged as navigable chains.
- Research synthesis: claims, evidence, counterevidence, competing explanations, and revision history in one inspectable structure.
- Organizational knowledge: decisions, dependencies, policy rationales, and change history extracted from document silos.
- AI retrieval and reasoning: structured context with versions, sources, premises, and opposition instead of undifferentiated text chunks.
- Public deliberation: disagreement attached to a concrete version and argument rather than a broad identity label.

10.2 The adoption constraint

The project's largest commercial risk is input cost. If users must manually construct every edge, the system will remain a specialist tool. If an AI agent can write directly into the public graph, quality will collapse. The scalable path is assisted structure: models extract candidate nodes, relationships, and duplicate warnings; people confirm intent; the protocol, not the model, decides admission.

Business model selection is intentionally deferred. Subscription, organizational deployment, developer access, or other revenue paths are only credible after the product proves that structured understanding and auditability create enough value to overcome contribution friction. There is no sound basis for assuming advertising or token economics at this stage.

11. State of the project

As of August 2026, Knowledge Ball is a working prototype with a converging protocol specification. The web application is the authoritative shared implementation; Android and Windows package the same TypeScript baseline. The current release baseline is v0.2.0. Cross-platform release reliability, mobile WebGL stability, database security, and deterministic layout remain active validation areas.

Workstream	Status	Evidence or next acceptance gate
Interactive prototype	Implemented	3D knowledge field, node detail, core interactions, web build/deploy, fixed system core, shared packaging baseline.
Identity and Energy foundation	Implemented / hardening	Supabase migrations, account model, ledger precision and conservation design; security and concurrency closure remain.
Claim protocol	Committed design	Stable entities, immutable versions, preflight, version-bound stake, atomic verdict and settlement, failed-candidate exit.
Deterministic spatial model	Committed design	5R occupancy, semantic shells, complex-chain-first placement, FCC snapping, canonical cellID, incremental stability.
Verifiable history	Next	Append-only accepted events, hash continuity, rebuildable projections, independent backup and export.
AI-assisted knowledge work	Research direction	Candidate extraction, conflict detection, retrieval, and explanation with human confirmation and protocol admission.

12. Roadmap and acceptance gates

Now: close the protocol loop

- Persist stable entities and immutable claim versions without parallel legacy paths.
- Make preflight, eligibility, stake, adjudication, settlement, and projection one traceable server-authoritative flow.
- Remove failed first-round candidates from the knowledge surface while retaining accounting integrity.
- Enforce the public/private boundary in schema, events, APIs, and rendering, not only in UI state.

Next: make space reliable

- Produce stable positions from the same input across devices.
- Limit an insertion to local movement and preserve spatial memory.
- Complete canonical cross-face cell addressing, chain focus, label LOD, and mobile performance budgets.

- Eliminate WebGL lifecycle leaks during detail views, submissions, route changes, and long sessions.

Then: make history portable and independently checkable

- Rebuild projections from accepted events and test backup restoration routinely.
- Publish human- and machine-readable exports with stable IDs and version relationships.
- Add source-quality and domain-eligibility layers before expanding into high-risk topics.
- Experiment with independent replicas and optional Merkle anchoring without making chain infrastructure a runtime dependency.

13. Four tests of viability

Test	Current assessment	What must be proven
Theoretical	Strong	Versioned claims, explicit reasoning, reversible verdicts, and public/private separation form a coherent model.
Engineering	Moderate	A bounded domain is feasible; deterministic 3D layout, mobile rendering, concurrent accounting, and relation editing are the hard parts.
Commercial	Unproven	Users or organizations must receive more value from structured understanding than they spend creating and reviewing structure.
Internet scale	Low to moderate	Requires Sybil resistance, domain governance, tiered storage, incremental layout, cold-history archival, and operating evidence not yet available.

This distinction matters. A model can be theoretically sound and still fail as a product. A prototype can run and still fail under adversarial incentives. A useful specialist tool can still fail at public-network scale. The roadmap is designed to test these layers in order rather than treating a compelling visualization as proof of the whole thesis.

14. Open governance questions

- How should domain eligibility combine graph distance, demonstrated calibration, professional evidence, and open participation?
- Which claim classes require higher thresholds, stronger sources, or named accountable reviewers?
- How should source independence and methodological quality be represented without collapsing them into citation count?
- How should value claims and predictions be displayed so they are not mistaken for settled facts?

- How can canonical English labels support cross-language de-duplication without erasing original wording or embedding linguistic bias?
- Which protocol parameters may the operator change during the prototype stage, and what proposal, test, notice, and rollback process should govern later changes?
- What event and proof formats would allow independent Knowledge Ball instances to verify or exchange claims without assuming a single global operator?

The credible answer is not to label the project decentralized before the power structure changes. Early operation is necessarily concentrated. Trust improves when authority is explicit, parameter changes are versioned, events are inspectable, failures can be rolled back, and independent parties can verify increasingly large parts of the system.

15. Closing

Knowledge Ball is an attempt to improve the unit of online knowledge. Instead of a page that can be overwritten, it offers a stable identity with versioned claims. Instead of a comment dispute, it offers explicit arguments tied to concrete conclusions. Instead of a single public score masquerading as personal understanding, it keeps shared validity and private mastery apart. Every knowledge node is a seam waiting to be worked: anyone can contribute what they know and draw Energy from the value that knowledge creates.

Commitment Every conclusion can be questioned. Every revision leaves a reason. Every person can see the real distance between themselves and what they want to know.

Appendix: current protocol parameters

Parameter	Current value	Meaning
Knowledge-ball diameter	R	Base visual and layout unit.
Knowledge-node target/exclusion distance	5R	FCC candidates within 5R of an occupied knowledge position are reserved.
Shell expansion increment	5R	A component that cannot fit expands outward in 5R steps.
Submission stake	1 Energy	Applies to a new or materially changed claim version.
Judgment stake	1 Energy	Each support or oppose action, bound to a specific version.
User balance floor	-10	Current design parameter; requires abuse controls.
Ledger precision	0.000001	Deterministic multi-party allocation.
Threshold sequence	1, 2, 4, 8, ...	Stepwise growth with eligible population; bands are versioned.
Authoritative spatial address	(shellID, cellID)	Canonicalized across icosahedral face boundaries.
Generation address	(shell, face, i, j, k)	Construction-only; not the persisted identity.
Product release baseline	v0.2.0	Prototype baseline as of August 2026.